NIKITA MALIK

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EDUCATION

Carnegie Mellon University, Masters in Computer Vision | GPA: **4.11/4.0** 2025 - Dec 2026
Coursework: Visual Learning and Recognition, Advanced Computer Vision, Introduction to Robot Learning
Manipal Institute of Technology, B.Tech in Information Technology | GPA : **9.4/10** 2021-2025

PROFESSIONAL EXPERIENCE

Machine Learning Engineering Intern - *Adobe* May 2026 - Present

- Developing an evaluation framework for LLM agents performing Natural Language-to-SQL conversion.

Graduate Research Assistant - *Aurora* Jan 2026 - May 2026

- Built a **closed-loop AD benchmark** for evaluating end-to-end planning policies in complex long-tail scenarios.
- Developed a rasterization pipeline to generate on-policy simulation rollouts for scalable planner training and evaluation.
- Benchmarked SOTA planning models on adversarially perturbed **nuPlan** environments and **Waymo Open Motion Dataset** identifying planner's failure modes and gaps.

Machine Learning Intern – *Lexsi Labs* Jan 2025 – July 2025

- Engineered a **post-training model alignment library** for relevance-based pruning using multiple attribution methods across CNNs, BERT, ViTs, and LLMs; achieved up to 65% model compression.

Computer Vision Research Intern - *Bosch Research* Nov 2023 - Oct 2024

- Developed an end-to-end **unsupervised optical flow estimation** pipeline for label transfer in autonomous vehicles.
- Implemented dataset interleaving scripts and custom evaluation methods proposed for heterogeneous camera setup.
- Attained a 20% improvement in end-point error over existing SOTA methods through joint optimization of flow estimation and domain adaptation tasks for **motion estimation in autonomous vehicles**.

RESEARCH EXPERIENCE

Research Assistant - *Human Sensing , CMU* Sept 2025 - Dec 2025

- Explored the use of **2D VLMs** for scene understanding through **multi-view reasoning** and view selection; integrated Gaussian splatting and depth-guided representations for enabling spatial and contextual understanding.

MITACS Research Intern - *University Of Calgary* May 2024 - Aug 2024

- Created a framework for identifying bias and applying various single-stage and multi-stage **bias mitigation** strategies; Contributed to Ischemic Stroke Segmentation Challenge; secured **rank 3 globally**; work presented at **MICCAI 2024**.

Research Intern - *IIT Hyderabad* May 2023 - May 2024

- Worked on enhancing fairness in ML models leveraging **meta learning** and **gradient alignment techniques**.
- Devised a novel MAML based algorithm to tackle sub-population bias. Recorded up to **20% improvement** in metrics.

SELECTED PUBLICATIONS

- *Nikita Malik* et al: FaAlGrad: Fairness through Alignment of Gradients across Different Subpopulations [Paper] (Published in Transactions of Machine Learning Research Journal (**TMLR Journal 2024**))
- *Nikita Malik* et al: Interpretability-Aware Pruning for Efficient Medical Image Analysis [Paper] (**MICCAI'25 WS**)

SKILLS

Programming Languages/Tools: Python, C++, Java, JavaScript, SQL, Docker
ML and CV Frameworks/Libraries: PyTorch, TensorFlow, OpenCV, Scikit-Learn, Matplotlib, Shap, Cuda
ML Topics: ML and DL, Transformers, LLMs, VLMs, Generative Models, Optical Flow, Scene Understanding

RELEVANT PROJECTS

- **Diffusion based semantic-prior module** [Code] [Slides] Developed a **diffusion-based module** to improve semantic segmentation results in adverse conditions. **Enhanced mIoU by 15%** on CityScapes dataset using Diffusion prior with RobustNet. Achieved comparable mIoU on DarkZurich and BDD 100K dataset.
- **Portfolio Generator** Developed mypage.cv, an automatic portfolio website builder that extracts structured data from resumes to generate and deploy personalized web pages with clean sub-routing. Leveraged **OpenAI APIs** for content generation and **Azure Blob Storage** for scalable file management.
- **Eye can do: For visually impaired people** [Code] [Cert.] Built a mobile application for visually impaired users with **facial recognition** linked to the phone's contact directory. Integrated **YOLO** for object detection and utilized camera images for **depth estimation**. Enabled real-time haptic alerts for safer navigation.

AWARDS AND ACHIEVEMENTS

- **DAAD WISE Scholarship 2024** Received scholarship to pursue research project at TU Munich.
- **Mitacs GRI 2024** A 12-week paid internship program accompanied by a 15,000 CAD scholarship.
- **Manipal Hackathon 2024** Ranked first among more than 400 teams winning a prize of 80,000 INR.